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Remember: Any group members who did **not contribute to the project should be given all zero (0) points for the collaboration grade on the GWP submission page.*

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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Step 1

Denoising removes the noise while keeping important information. The noise (random changes) can prevent the signals (real changes) being correctly detected, and removing the noise enhances the signals.

Features:

- Separating signal and noise**:** it can distinguish meaningful signals from random noise, often based on the assumptions that noise would follow a normal distribution.
- Learning meaningful patterns: it identifies the underlying structure of the data and learns to ignore the random noise.
- The Marchenko-Pastur: it works by replacing the eigenvalues classified as random noise by Marčenko-Pastur with a constant eigenvalue in covariance matrix. This makes true signals stronger even if they are small. It is particularly useful when the dataset is small and contains more random noise and weaker signals(Timm, 2021).

Benefits:

- Improves accuracy: it helps learning meaningful patterns and ignore noise and allows the model to make more accurate and reliable predictions, reducing overfitting and increasing generalisability.
- Clear view and better decision making: it provides a clearer view of what is truly happening and better supports decisions based on genuine patterns. It makes the true patterns in data much easier to see and interpret. This helps to make better decisions.
- Reduces computation needs: Cleaner data also increases the processing efficiency and reduces the computation cost and storage needs.

Clustering is a method that groups similar data points together. It does not need pre-existing labels. It allows exploring the hidden patterns in data. It is widely used in finance, marketing, health, and social media.

Features:

- K-means algorithm: an unsupervised learning algorithm that splits data into a number of groups (K) based on distance to a centre point (centroids). The centroids are initially randomly chosen, so k-means should be run multiple times to ensure the optimal results. It is suitable for large datasets.
- Hierarchical Clustering: it creates a tree of groups that can be divided at different levels. Therefore, it works better with a dataset that can be transformed into a hierarchical structure (Google Developers, 2015).

Benefits:

- Hidden Patterns: It can group data points based on underlying patterns that are not obvious, and offers new insights.
- Simplifying Complex Data: it groups large data into meaningful clusters, so it becomes easier to summarise, visualise and interpret.
- Outlier: It helps identify data points that are significantly different from the rest of the data. It is useful in error detection (Google Developers, 2015).

Backtesting is a key method in finance. It lets traders and investors test a strategy using past market data. By applying a strategy to historical data, they can check the model's reliability and profitability (Lopez de Prado and Lewis, 2018).

Key Features

- Simulation: it allows simulation with historical data to test how a strategy would have performed.
- Performance testing: it allows testing trading strategies against different investment objectives and scenarios before real investment. It enables tuning of model parameters such as weights and threshold to assess their impact.
- Cross-Validation Integration: the use of advanced techniques like Combinatorial Purged Cross-Validation (CPCV) can reduce overfitting and increase generalisability (Berger, 2019).

Benefits

- Risk assessment: it allows examining the simulated performance to identify potential risks, drawdowns and underperformance.
- Model improvement: it allows adjusting different parameters and testing under different market conditions to improve the trading strategy.
- Confidence: simulating a portfolio strategy before investment can increase the investor's confidence in the trading strategy.

Step 2

The best portfolio from GWP2 was the Black-Litterman(BL) portfolio that had the lowest sharpe ratio. We will be applying the following improvements to the portfolio:

Single improvement:

1. Denoising: removes random noise from the covariance matrix and enhances the meaningful signals. It will help increase the risk reward relationship, i.e. sharp ratio.
2. Clustering: groups similar assets based on correlations and increases diversification. It will help reduce the portfolio volatility and max drawdown.
3. Backtesting: evaluates portfolio performance using historical data to test robustness. It will help reduce overfitting and improve generalisability.

Combined improvements:

4. Denoisng + Clustering: enhances signals and increases diversification
5. Denoisng + Clustering + Backtesting: builds on top of the previous improvement, and backtests with historical data to reduce overfitting.

Step 3 & Step 4

The training data used was the exact training dataset used in GWP2. It consisted of daily stock prices of 5 assets (Apple, Google, Microsoft, Amazon and NVIDIA) from 2023-01-01 to 2023-06-31. The 1-month test period was from 2023-07-01 to 2023-07-30.

We performed the three different improvements individually, and the combinations of two or even three methods together to test the performance. The original Black-Litterman(BL) model was used as a base bench mark. Their performance metrics in the out-of-sample test are summarised below.

Table 1: Weight Summary across Models

	Base_BL	**BL_Denoised**	**BL_Clustered**	**BL_Denoised _Clustered**	**BL_Backtested _Average**	**All_Three _Average**
AAPL	0.6437	0.6529	0.7453	0.7499	0.6437	0.6407
MSFT	0.0107	0.0000	0.0132	0.0183	0.0107	0.0444
GOOG	0.0000	0.0000	0.0000	0.0000	0.0000	0.0293
AMZN	0.0000	0.0000	0.0106	0.0000	0.0000	0.0336
NVDA	0.3456	0.3471	0.2309	0.2319	0.3456	0.2518

Figure 1: Portfolio Weights Comparison

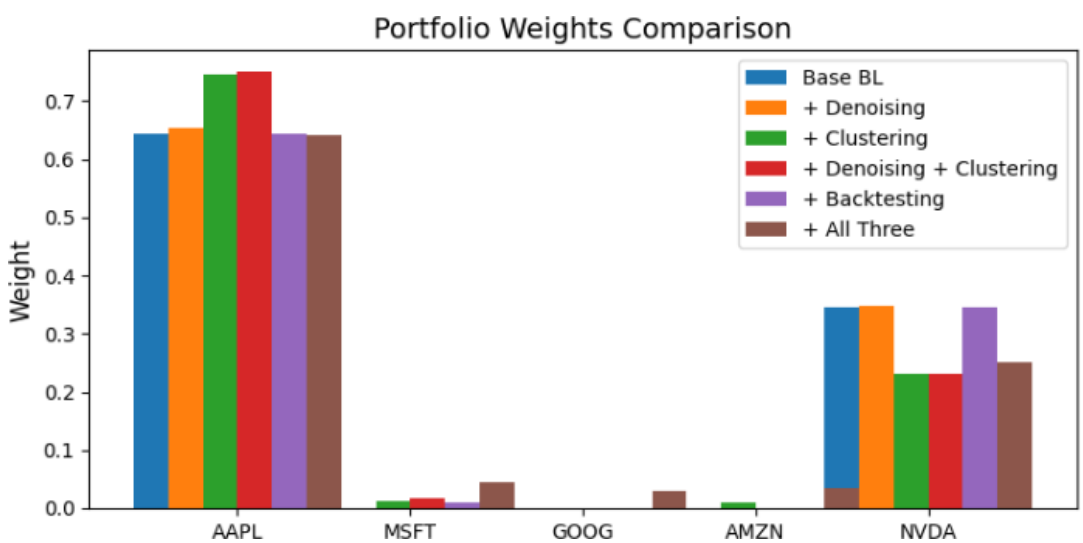

Table 2: In-Sample 6-month Performance Comparison:

Portfolio	Cumulative Return	Volatility	Sharpe Ratio	Sortino Ratio	Max Drawdown	Expected Shortfall (95%)
Base_BL	91.63%	28.32%	4.767	24.772	-6.13%	-2.26%
BL_Denoised	91.83%	28.37%	4.768	24.908	-6.11%	-2.26%
BL_Clustered	77.64%	24.56%	4.817	23.394	-5.59%	-2.03%
BL_Denoised_Clustered	77.89%	24.58%	4.825	23.465	-5.59%	-2.03%
BL_Backtested_Average	91.63%	28.32%	4.767	24.772	-6.13%	-2.26%
All_Three_Average	79.20%	25.40%	4.739	21.991	-5.82%	-2.11%

Table 3: Out-of-Sample 1-month Performance Comparison:

Portfolio	Cumulative Return	Volatility	Sharpe Ratio	Sortino Ratio	Max Drawdown	Expected Shortfall (95%)
Base_BL	4.67%	16.74%	3.688	8.571	-3.13%	-1.84%
BL_Denoised	4.68%	16.70%	3.707	8.700	-3.11%	-1.81%
BL_Clustered	3.68%	14.80%	3.254	7.629	-2.66%	-1.59%
BL_Denoised_Clustered	3.71%	14.78%	3.279	7.646	-2.66%	-1.60%
BL_Backtested_Average	4.67%	16.74%	3.688	8.571	-3.13%	-1.84%
All_Three_Average	4.08%	15.66%	3.424	7.552	-2.87%	-1.81%

Step 5.1 Out-of-Sample Performance Comparison

1. Single improvement method

When using a single improvement on its own, denoising was the best-performing method (cumulative return: 4.68% and sharpe ratio: 3.7) when compared to the base black-litterman (BL) portfolio and the other two methods, clustering and backtesting. However, denoising as an improvement method did not

significantly outperform the base BL portfolio, as their performance was very similar. When backtesting was used on its own, it did not improve the performance, and its average performance was almost identical to the original base BL portfolio. Clustering showed the worst return performance among all models, with cumulative return of 3.68% and sharp ratio of 3.25. However, clustering also showed a lower volatility, max drawdom and expected shortfall, and these indicate lower risks in the portfolio. This demonstrates the classic trade-off between risk reduction and return maximisation

2. Multiple Improvement methods

The combination of all three improvement portfolios (cumulative return: 4.08%, sharp ratio: 3.42) achieved a slightly higher return than the combination of denoising and clustering (cumulative return: 3.71%, sharp ratio: 3.28). However, the combination of denoising and clustering portfolio showed better risk management with lower values of risk metrics. The portfolio that combined all three improvements produced the lowest Sortino Ratio. This means it controls downside risk at the cost of returns. Overall, none of the portfolios that used combined improvements outperformed the base BL or single improvement portfolios on sharp ratio.

Step 5.2 Reasons for the performance

The difference in the performance can be summarised below:

1. Denoising improves return performance and better return to risk ratio
2. Clustering reduces risks at the cost of return
3. None of the single improvement methods or their combination significantly improves the prediction performance.

Using denoising showed slight improvement in return metrics is expected, because it is intended to detect signals and remove noise. In contrast, the clustering is intended for grouping assets and increases diversification for better risk management. This could be seen in the weight distribution for the 5 assets: portfolios using clustering have incorporated at least 4 out of 5 assets, whereas the portfolio with denoising only holds 2 assets, Apple and NVIDIA, and is more susceptible to short-term price fluctuations. For example, a shortage of chip material supply or an issue with transport would have a more direct and bigger impact on Apple and NVIDIA directly. Therefore, the denoising portfolio had the largest max drawdown of -3.13%, compared to -2.66% in the clusterings portfolio.

It was surprising that none of the improvements significantly improves the efficiency of the original portfolio when we looked at the cumulative return and sharpe ratio. There could be many reasons to this:

1. Financial products: we chose a few of the best performing technology assets. During the training and testing period (all in 2023), the market is in the middle of the AI boom. These stocks have already been reported to be at their historical highs. As a result, additional improvements are unlikely to create further efficiency improvement. With denoising that is known to increase the model's sensitivity to signal, its improvement was very limited because the true signals were already much stronger than the noise. Future research can look at replacing a few assets from a different sector to better evaluate the impact of different improvement methods.

2. Testing period: The testing period of 1-month was too short. It was only for 1 month straight after the training period. As the market is still experiencing the AI boom, and the technology assets are expected to continue to grow. It would be beneficial to extend the testing period or use pre-2023 data to see if the performance of the out-of-sample stands. Furthermore, the benefit of diversification from clustering would be more evident when there is more uncertainty or frequent fluctuations in the market.

Step 5.3 Justification the use of improvement methods

From the perspective of incremental gain, the performance results do not justify the additional complexity and effort of the improvement. However, as previously discussed, the analysis was completed during the AI bloom using popular technology assets, and the strong performance of the technology market might have undermined the effectiveness of the improvement methods.

On the other hand, the combination of all three improvements created a more balanced portfolio with relatively higher return, lower risk and greater diversification. It was the only portfolio that incorporated all 5 assets. The increased diversification did not harm its' cumulative return, and it outperformed the combination of denoising and clustering that only selected 4 assets and has slightly less diversification.

Therefore, the project could not fully conclude whether the additional complexity is justified, due to the limitations of chosen financial product and data period. Future research could aim to better test the effectiveness of the improvement methods. Nevertheless, the project demonstrates the different improvements impact performance metrics differently, and they can accommodate investors with varying risk appetite. For example, a risk-averse investor could benefit from clustering that helps reduce portfolio volatility, and those who are more interested in high return can benefit from denoising. Using all three different improvements can achieve a more balanced portfolio to enhance diversification and risk management, while persevering considerably higher returns.

Step 6

1. Overcoming Estimation Error in Portfolio Management

- Traditional portfolio optimization methods, like the Nobel Prize-winning Mean-Variance Optimization (MVO), are critically flawed by their sensitivity to estimation error.
- Estimation error arises when we use limited historical data to estimate a portfolio's future risks and returns.
- It occurs when estimated (sample) parameters (expected returns, covariances) are different from their true but unknown values (population parameters):

$$\hat{\mu} \neq \mu \quad \text{and} \quad \hat{\Sigma} \neq \Sigma$$

- For example, portfolio variance is defined as

$$\text{Portfolio Variance} = w^T * \Sigma * w$$

When our estimate of Σ is estimated with errors, our final portfolio allocation becomes unreliable.

2. Estimation Error in Our Data

- 5 technology asset: Apple, Microsoft, Amazon, Google and NVIDIA
- 6-month of daily returns from Jan 2023 to June 2023 - AI Boom
- Problem:
  - Small sample size - noisy covariance matrix
  - Strong growing trends - expected returns overestimated
  - Skewed allocation - initial heavy concentrated on Apple (64%) and NVIDIA (35%)
- Overly optimistic expectations, limited diversification and not sustainable

3. Improvement Methods

- Denoising
  - Removes random noise in covariance matrix using Marčenko–Pastur filtering
  - Enhances meaningful signals
- Clustering
  - Groups correlated assets using hierarchical clustering
  - Increases diversification for better risk management by ensuring weights are more spread across asset groups
- Backtesting
  - Runs multiple simulations to detect overfitting to the training sample
  - Increases generalisability of models / strategies
  - Did not directly resolve estimation error

4. Results and Connection to Estimation Error

- In-sample training
  - Denoising slightly improved return metrics (better signal detection).
  - Clustering reduced volatility and drawdowns.
- Out-of-sample testing
  - Both methods showed marginal benefits.
  - Improvement was limited by strong tech-sector bias and strong technology market growth.
- Key links
  - Denoising directly addressed covariance estimation error.
  - Clustering mitigated estimation error via diversification.
  - Backtesting validated robustness.

5. Key Takeaways

- Estimation error is unavoidable with small, noisy datasets.
- Denoising and clustering provided different approaches to reduce the impact of Estimation Error.
- Backtesting added confidence but did not fix estimation error itself.
- For investors: Choice of method depends on risk appetite → risk-averse (clustering), return-seeking (denoising), balanced (all three).

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